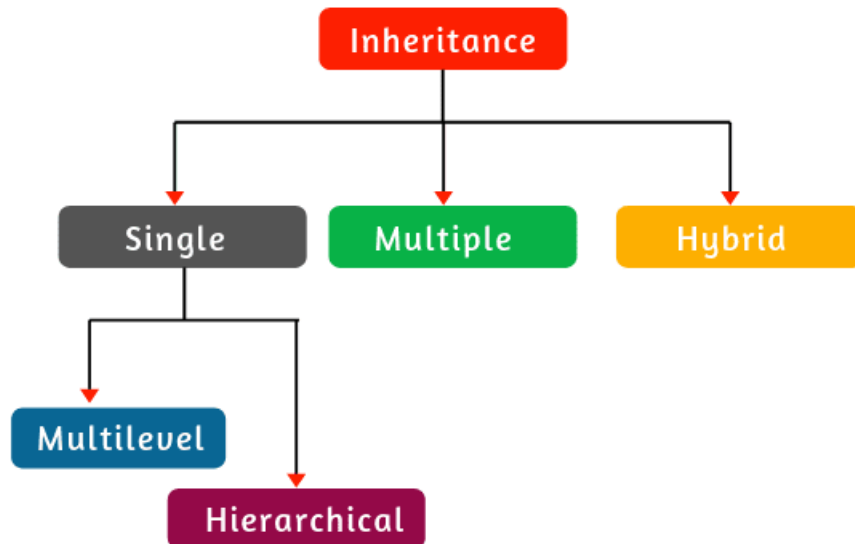


OOPS Concept

Types of Inheritance :-

- Java supports the following main types of inheritance:



2. Hierarchical Inheritance : –

- One parent class is inherited by two or more child classes.
- The parent class contains common properties and methods.
- Child classes inherit the parent's features and can have their own unique features.
- It provides code reusability by avoiding duplication of common code.
- Child classes are independent of each other, connected only through the parent.

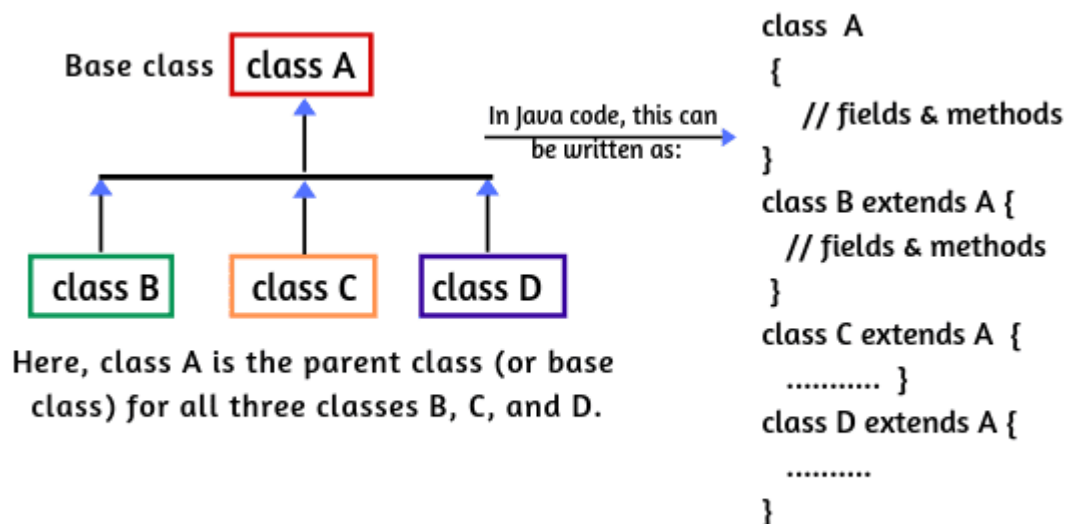


Fig: Hierarchical Inheritance in Java

Example –

```
package oops;

class parent1
{
    void p1()
    {
        System.out.println("I am Parent Class");
    }
}

class child1 extends parent1
{
    void c1()
    {
        System.out.println("I am Child Class 1");
    }
}

class child2 extends parent1
{
    void c2()
    {
        System.out.println("I am Child Class 2");
    }
}

public class hierarchical_inheritance extends parent1
{
    void c3()
    {
        System.out.println("I am Child Class 3");
    }

    public static void main(String[] args) {
        hierarchical_inheritance obj=new hierarchical_inheritance();
        obj.p1();
        obj.c3();
        System.out.println("-----");

        //child2 object
        child2 obj2 = new child2();
        obj2.p1();
        obj2.c2();
        System.out.println("-----");

        //child1 object
        child1 obj3 = new child1();
        obj3.p1();
        obj3.c1();
    }
}
```

Output –

I am Parent Class

I am Child Class 3

I am Parent Class

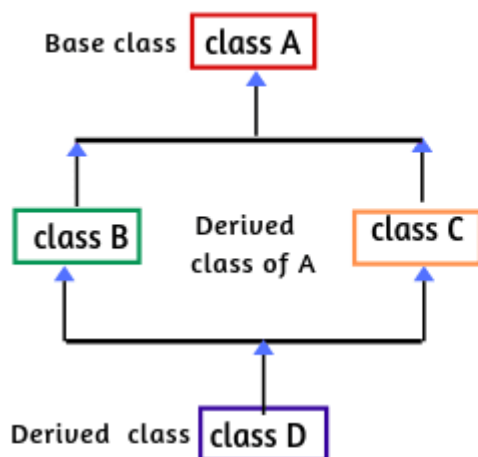
I am Child Class 2

I am Parent Class

I am Child Class 1

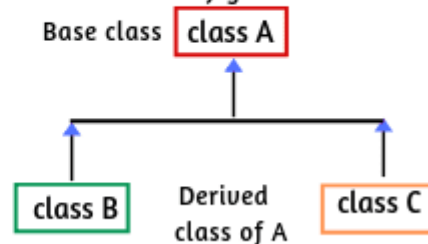
3. Hybrid Inheritance :-

- Hybrid inheritance combines two or more types of inheritance in a single program.
- In Java, it is implemented using classes and interfaces.
- It allows flexible and reusable code by combining features of different inheritance types.
- They Can be complex to manage in large programs.

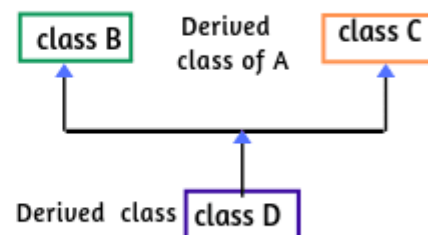


Combining together them as shown in fig , we have a hybrid inheritance.

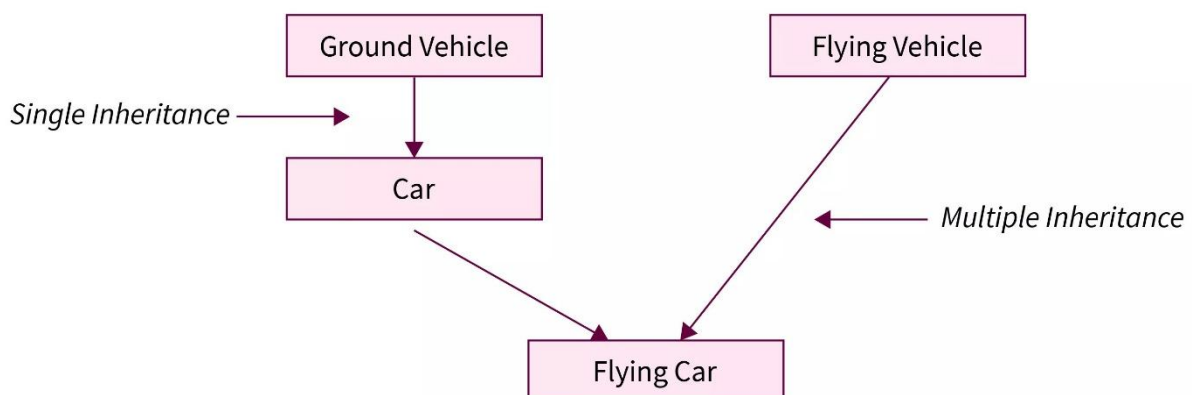
For the first half of the fig, we have hierarchical inheritance as shown by breaking the fig.



In second half, we have multiple inheritance as shown in the fig.



www.scientecheasy.com Fig: Hybrid Inheritance in Java



Example –

```
package oops;
class hy_parent
{
    void hp()
    {
        System.out.println("I am Parent Class");
    }
}
class hy_child1 extends hy_parent
{
    void hc1()
    {
        System.out.println("I am Child Class1");
    }
}
class hy_child2 extends hy_parent
{
    void hc2()
    {
        System.out.println("I am Child Class2");
    }
}

public class hybrid_inheritance extends hy_child2{
    void hsc()
    {
        System.out.println("I am Sub Child");
    }

    public static void main(String[] args) {
        hybrid_inheritance ob=new hybrid_inheritance();
        //for sub child
        ob.hsc();
        ob.hc2();
        ob.hp();
        System.out.println("-----");

        //for child2
        hy_child2 ob2=new hy_child2();
        ob2.hc2();
        ob2.hp();
        System.out.println("-----");

        //for child1
        hy_child1 ob1=new hy_child1();
        ob1.hc1();
        ob1.hp();
    }
}
```

Output –

I am Sub Child

I am Child Class2

I am Parent Class

I am Child Class2

I am Parent Class

I am Child Class1

I am Parent Class